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WEB APPENDIX

Comparative Myology of the Dog

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
FORELIMB						
Trunk to Forelimb						
Cutaneous trunci	No correlate; however, like the platysma in the neck	None: Twitch the skin over the scapular and shoulder area	In the superficial fascia of the dorsal, lateral, and ventral walls of thorax and abdomen	Medial side of the forelimb		Lateral thoracic nerve
Trapezius						
Cervical	Upper trapezius	Elevate the scapula cranially and dorsally; abduct the forelimb	Median or mid-dorsal raphe of the neck and supraspinous ligament	Dorsal aspect of the spine of the scapula	Caudoventral	Accessory nerve Cranial XI
Thoracic	Lower trapezius	Depress the scapula; abduct the forelimb	Supraspinous ligament and dorsal spinous processes of T3-T8 or T9	Middle part of the spine of the scapula	Cranioventral	Accessory nerve Cranial XI
The fibrous band in between the cervical and thoracic parts is in the position of the middle trapezius.						
Rhomboideus						
		Elevate the scapula and forelimb; pull the scapula to the trunk				Ventral rami of cervical spinal nerves
Rhomboideus capitis: two parts		Elevate the scapula and forelimb; pull the scapula to the trunk	Nuchal crest of the occipital bone	Cranial dorsal border of the scapula	Craniodorsal	
Rhomboideus cervicis	Rhomboid minor	Elevate the scapula and forelimb; pull the scapula to the trunk	Median fibrous raphe of neck, spinous processes of T1-T3	Caudal (base) of the scapula, medially and laterally	Craniodorsal	
Rhomboideus thoracis	Rhomboid major	Joint action in the forelimb	Spinous process of T4-T7	Caudal (base of scapula, medially and laterally)	Craniodorsal	

Serratus ventralis	Serratus anterior	Carry the trunk and shoulder forward and back, support the trunk, depress the scapula	Transverse processes of C3-C7 and ventral ribs 1-7 or 8	Medial scapula, dorsal one third	Craniodorsal and cranioventral	Ventral rami of cervical spinal nerves and long thoracic nerve
Brachiocephalicus: two parts	Two parts are divided by a fibrous clavicular intersection					Accessory nerve
Cleidobrachialis	Anterior deltoid	Extend the shoulder; advance the forelimb	Ventral aspect of the clavicle	Cranial surface, distal third of the humerus, between the biceps brachii medially and the brachialis laterally	Distomedial	Ventral rami of cervical spinal nerves
Cleidocephalicus—neck muscle		Draw the head to the side; extend the neck	Dorsal aspect of the clavicle	Cranial half middorsal fibrous raphe and nuchal crest of occipital bone	Dorsocranial	
Superficial pectoral: two parts—						
Descending pectoral portion	Part of pectoralis major	Adduct the shoulder, prevent abduction in weight-bearing; extend the shoulder; draw trunk sideward	Ventral sternum, most cranial aspect (ventral to the transverse pectoral muscle)	All but most distal part of crest of greater tubercle of humerus	Distolateral	Cranial pectoral nerves
Transverse pectoral portion	Part of pectoralis major	As above, plus flex or extend shoulder, depending on position	Ventral sternum, cranial half dorsal to the descending pectoral muscle	All but most distal part of crest of greater tubercle of humerus	Distolateral	Cranial pectoral nerves
Deep pectoral	Part of pectoralis major	Flex the shoulder; move the trunk cranially during weight-bearing; support the trunk with the serratus ventralis	Ventral aspect of the sternum, dorsal to the superficial pectoral muscle cranially	Major portion—lesser tubercle of humerus, aponeurosis to the greater tubercle and crest of humerus	Craniodistolateral	Caudal pectoral nerves
				Caudal portion—medial brachial fascia		

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
Latissimus dorsi	Latissimus dorsi	Flex the shoulder; draw the trunk cranially in weight bearing; adduct the shoulder; support the trunk	Thoracolumbar fascia from spinous processes of most caudal seven or eight thoracic and lumbar vertebrae; most caudal two or three ribs	Teres major tuberosity on the humerus and teres major tendon; joins tip of the muscle cutaneous trunci laterally	Craniolatero-distal	Thoracodorsal nerve
SHOULDER: WITHIN FORELIMB						
Deltoideus: two parts	Deltoid			The two parts of the deltoid blend near the shoulder joint and attach to the deltoid tuberosity		
Spinous head	Posterior deltoid	Shoulder flexion	Scapular spine, blending with the infraspinatus		Distolateral	Axillary nerve
Acromial head	Middle deltoid	Shoulder abduction	Acromion		Laterodistal	Axillary nerve
Infraspinatus	Infraspinatus	Shoulder flexion or extension depending on position; shoulder abduction and lateral rotation; stabilize lateral shoulder	Infraspinous fossa	Lateral side of the greater tubercle of the humerus	Laterodistal Laterodistal	Suprascapular nerve
Teres minor	Teres minor	Shoulder flexion and lateral rotation	Infraglenoid tubercle	Infraglenoid tuberosity of the humerus	Laterodistal	Axillary nerve
Supraspinatus	Supraspinatus	Extend the shoulder; stabilizes and prevents collapse of the shoulder	Supraspinous fossa, wider and larger than the infraspinatus	Greater tubercle of the humerus by a thick tendon	Distally on the cranial aspect of the shoulder, then medially	Suprascapular nerve
Subscapularis	Subscapularis	Adduct, extends, and stabilizes the shoulder medially	Subscapular fossa	Lesser tubercle of the humerus	Craniodistal	Subscapular nerve

Teres major	Teres major	Shoulder flexion and medial rotation	Caudal angle and border of the scapula and caudal subscapularis	Teres major tuberosity of the humerus	Cranial and then distal	Axillary nerve
Coracobrachialis	Coracobrachialis	Shoulder adduction and extension	Coracoid process of the scapula	Crest of the lesser tubercle of humerus, proximal to teres major tuberosity	Distolateral	Musculocutaneous nerve
Biceps brachii—one head	Biceps brachii	Shoulder extension and elbow flexion; some passive stability in weight bearing, preventing shoulder flexion	Supraglenoid tubercle of the scapula	Ulnar tuberosity on proximal cranial ulnar	Distomedial	Musculocutaneous nerve
Brachialis	Brachialis	Elbow flexion	Proximal third of the lateral humerus (brachialis groove)	Radial tuberosity on proximal cranial radius (weaker part)		
Triceps brachii—four heads	Triceps brachii—three heads	Elbow extension; helps tense the antebrachial fascia		Ulnar tuberosity on proximal cranial ulnar	Distolateral	Musculocutaneous nerve
Long head	Long head	Elbow extension, shoulder flexion	Caudal border of the scapula	Olecranon tuber or tuberosity	Caudodistal	Radial nerve
Lateral head	Lateral head	Elbow extension	Tricipital line of the humerus—covers the accessory head	Olecranon tuber, lateral to long head	Caudodistal	
Accessory head	None	Elbow extension	Neck of the humerus	Olecranon tuber, between lateral and medial heads	Caudodistal	
Medial head	Medial head	Elbow extension	Crest of lesser tubercle near teres major tuberosity	Olecranon	Caudodistal	
Anconeus	Anconeus	Elbow extension; helps tense the antebrachial fascia	Lateral supracondylar crest; lateral and part of medial epicondyles of the humerus	Lateral surface, proximal end of ulna	Caudal	Radial nerve

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
Tensor fasciae antebrachii	None	Elbow extension; chief tensor of the antebrachial fascia	Fascia covering the latissimus dorsi; very thin	Olecranon and the medial fascia of the forearm	Distolateral	Radial nerve
CRANIAL AND LATERAL FOREARM OR ANTEBRACHIUM						
Brachioradialis	Brachioradialis	Supination; rotation of the radius dorsolaterally	Lateral supracondylar crest of the humerus; thin and inconstant muscle	Distal fourth of the radius	Medial then distal	Radial nerve
Extensor carpi radialis	Extensor carpi radialis longus and brevis	Carpal extension; elbow flexion	Lateral supracondylar crest	Small tuberosities on dorsal bases of MCs II and III		Radial nerve
Common digital extensor	Extensor digitorum communis	Extension of digits II-V; carpal extension	Lateral epicondyle of the humerus	Extensor processes of the distal phalanges of digits II-V	Distomedial	Radial nerve
Lateral digital extensor	None	Extension of digits III-V; carpal extension	Lateral epicondyle of the humerus	Proximal ends, all phalanges III-V; extensor processes of distal phalanges of III-V	Distomedial	Radial nerve
Ulnaris lateralis	Extensor carpi ulnaris	Carpal ulnar deviation or abduction; weak carpal flexion	Lateral epicondyle of the humerus	Lateral aspect, MC V; accessory carpal bone	Distolateral	Radial nerve
Supinator	Supinator	Supination; elbow flexion	Lateral epicondyle of the humerus	Cranial surface, proximal fourth of the radius	Distomedial	Radial nerve
CAUDAL AND MEDIAL FOREARM OR ANTEBRACHIUM						
Pronator teres	Pronator teres	Pronation, elbow flexion	Medial epicondyle of the humerus	Medial border of the radius between middle and proximal thirds	Distolateral	Median nerve

Flexor carpi ulnaris: two parts	Flexor carpi ulnaris	Carpal flexion, carpal abduction	Proximally the ulnar head is more cranial and distally the humeral head is more cranial	Ulnar nerve
Ulnar head			Caudal border and medial surface of the olecranon	
Humeral head			Accessory carpal bone	
Flexor carpi radialis	Flexor carpi radialis	Carpal flexion	Medial epicondyle of the humerus	Median nerve
Superficial digital flexor	Flexor digitorum superficialis	Flexion of PIP and MCP of II-V; carpal flexion	Medial epicondyle of the humerus and medial border of radius	Median nerve
Deep digital flexor: three parts	Flexor digitorum profundus	Flexion of digits; carpal flexion	Medial epicondyle of the humerus—radial, deep, and medial bellies	Median nerve to radial head, deep and medial parts of humeral head
Humeral head			For all three heads: palmar surface of the base (proximal end) of the distal phalanx of digits II-V and a weak tendon to digit I	
Ulnar head	None		Proximal three fourths of caudal border of the ulna	Ulnar nerve to ulnar head and lateral part of humeral head
Radial head	None		Middle third, medial border of the radius	
Pronator quadratus	Pronator quadratus	Pronation	Medial surface of the ulna, caudal to the interosseous membrane	Median nerve
			Caudal surface of the radius except proximal and distal ends	
			Distomedial	

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
HINDLIMB						
Caudal Thigh Muscles						
Biceps femoris: two or three heads—longest and widest thigh muscle in the hindlimb	Biceps femoris	Cranial—hip and stifle extension Caudal—stifle flexion, tarsocrural extension, hip (hindlimb) abduction	Sacro-tuberous ligament and ischiatic tuberosity	Patella; patellar ligament and cranial border of tibia via fascia lata and crural fascia; tibial body via crural fascia; calcaneal tuberosity	Distocranio-lateral	Sciatic nerve
Caudal crural abductor muscle	None	Stifle flexion and hip (hindlimb) abduction with caudal biceps femoris muscle	Sacro-tuberous ligament near ischiatic tuberosity	Via crural fascia to the digital extensor	Distocranio-lateral	Sciatic nerve
Semitendinosus	Semitendinosus	Hip and tarsocrural extension; stifle flexion	Ischiatic tuberosity	Medial tibial body and calcaneal tuberosity via crural fascia	Distocranio-medial	Sciatic nerve
Semimembranosus: two bellies	Semimembranosus	Hip extension Femoral belly—stifle extension Tibial belly—stifle flexion or extension, depending on stifle position	Ischiatic tuberosity	Medial lip, distal shaft, caudal rough surface of the femur—distal to pectineus muscle; medial aspect, proximal part, medial tibial condyle	Distocranio-medial	Sciatic nerve
MEDIAL THIGH MUSCLES						
Sartorius	Sartorius					
Cranial part		Hip flexion, stifle extension	Iliac crest, thoracolumbar fascia	Patella, with rectus femoris	Distocaudal	Femoral nerve
Caudal part		Hip flexion, stifle flexion	Cranial VIS, adjacent ventral ilium	Cranial border, medial tibia, with gracilis		
Pectineus	Pectineus	Hip or hindlimb adduction	Body of pubis, iliopubic eminence to pubic tubercle	Medial lip, distal shaft, caudal rough surface of the femur, proximal to semimembranosus		Obturator nerve

Adductor: two parts, which may or may not be distinguishable	Adductors magnus, brevis, and longus	Hip or hindlimb adduction, hip extension	Entire symphysis pelvis via symphyseal tendon, adjacent ischiatic arch, ventral pubis and ischium	Entire lateral lip of the caudal rough surface of the femur	Distocranio-lateral	Obturator nerve
LATERAL THIGH MUSCLES						
Tensor fasciae latae	Tensor fascia lata/tensor fasciae latae	Tense the fascia lata, hip flexion, stifle extension	Cranial VIS and adjacent ilium (=tuber coxae), aponeurosis of middle gluteal muscle	Via fascia lata, connects to biceps femoris and quadriceps femoris muscle to patella	Distocaudal	Cranial gluteal nerve
Superficial gluteal	Gluteus maximus	Hip extension, hip or hindlimb abduction	Lateral border of sacrum, caudal 1 or 1st vertebra of the tail, sacrotuberous ligament, cranial DIS via deep gluteal fascia	Third trochanter, fusing with aponeurosis of tensor fasciae latae—synovial bursa deep to attachment in one third of dogs	Caudolatero-distal	Caudal gluteal nerve
Middle gluteal	Gluteus medius	Hip extension, abduction, internal rotation	Crest and gluteal or lateral surface of the ilium	Greater or major trochanter	Caudolatero-distal	Cranial gluteal nerve
Piriformis—may actually be part of the middle gluteal or deep gluteal muscle	Piriformis	Hip extension	Lateral surface of S3 and C1	Greater trochanter with middle gluteal muscle	Caudolatero-distal	Caudal gluteal nerve
Deep gluteal	Gluteus minimus	Hip extension, abduction, internal rotation	Body of ilium, ischiatic spine	Greater trochanter, cranial aspect	Caudodistal	Cranial gluteal nerve
CRANIAL THIGH MUSCLES						
Iliopsoas	Iliopsoas	Hip flexion				Ventral rami of lumbar and femoral nerves

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
Psoas major	Psoas major	Hip flexion, vertebral column flexion, lumbar spine extension with ventral tilting of the pelvis	Transverse processes and bodies of the lumbar vertebrae	Lesser trochanter with the iliacus	Caudoventro-lateral	
Iliacus	Iliacus	Hip flexion	Smooth ventral surface of the ilium	Lesser trochanter with the psoas major	Caudoventro-lateral	
Quadriceps femoris	Quadriceps femoris	See actions below				
Rectus femoris	Rectus femoris	Stifle extension, hip flexion	Lateral ilium, cranial to the acetabulum—bursa occasionally deep to this	Tibial tuberosity	Distocranio-lateral	Femoral nerve
Vastus lateralis	Vastus lateralis	Stifle extension	Proximal part of the lateral lip of the caudal rough surface of the femur	Tibial tuberosity—bursa usually deep to tendon attachment	Distocranio-medial	Femoral nerve
Vastus intermedius	Vastus intermedius	Stifle extension	Lateral side of the proximal femur, with the vastus lateralis	Tibial tuberosity	Distocranio-medial	Femoral nerve
CRANIAL THIGH MUSCLES						
Vastus medialis	Vastus medialis	Stifle extension	Medial side, proximal end of the cranial end, medial lip, caudal rough surface of the femur	Tibial tuberosity—bursa usually deep femur; proximal attachment	Distocranio-lateral to tendon	Femoral nerve
Articularis genu	Articularis genu	Stifle extension; tenses the proximal pouch of the stifle joint capsule	Cranial surface of femur, proximal to the trochlea	Tibial tuberosity		Femoral nerve

CRANIO-LATERAL MUSCLES OF THE LEG OR CRUS

Cranial tibial	Anterior tibialis/ tibialis anterior	Talocrural flexion; rotates the hindpaw so the plantar surface of the paw faces medially	Extensor groove and adjacent articular margin of tibia, lateral edge of the cranial border of the tibia	Plantar surface of base of MT I and II after coursing deep to the crural extensor retinaculum	Distomedial	Deep peroneal nerve
Long digital extensor	Extensor digitorum longus	DIP and PIP extension II-V, talocrural flexion	Extensor fossa on the femur lateral to the trochlea	Extensor processes on the dorsum, distal phalanges of digits II-V coursing deep to the crural and tarsal extensor retinacula	Distomedial	Peroneal nerve
Extensor digiti I longus	Extensor hallucis longus and part of the extensor digitorum longus	Digit II extension, digit I extension	Cranial border of the fibula between proximal and middle third of the interosseous membrane cranial to the peroneus brevis	Fascia and tendon of the lateral digital extensor of MTP joint II; aponeurosis of the metatarsus, rudiment of digit I or hallux	Distomedial	Deep peroneal nerve
Lateral digital extensor	Peroneus tertius	Digit V extension and abduction	Proximal third of the fibula	Extensor process on the dorsum of middle phalanx of digits V coursing deep to the crural and tarsal extensor retinacula		Superficial peroneal nerve
Peroneus longus	Peroneus longus	Talocrural flexion, rotates the hindpaw so the plantar surface of the paw faces laterally	Lateral tibial condyle, proximal fibula, lateral collateral ligament of the stifle, which then attaches to the lateral femoral epicondyle	Fourth tarsal bone/ quartal bone, plantar aspect of the base of the MTs after coursing deep to the abductor digiti V muscle	Distal on the leg then medial on the plantar hindpaw	Common peroneal nerve, deep peroneal nerve
Peroneus brevis	Peroneus brevis	Talocrural flexion	Lateral surface of the distal two thirds of the fibula and tibia	Proximal end of MT V in a common sheath with the extensor digitorum lateralis after coursing deep to the lateral collateral ligament of the ankle	Distal and slightly medial	Superficial peroneal nerve

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
CAUDAL MUSCLES OF THE LEG						
Popliteus	Popliteus	Stifle internal rotation/ leg internal rotation	Lateral condyle of the femur	Proximal third caudal surface of tibia, with a sesamoid at the muscle-tendon junction	Distomedial	Tibial nerve
Gastrocnemius: two heads	Gastrocnemius—two heads	Talocrural extension; stifle flexion	Medial and lateral supracondylar tuberosities of the femur; each tendon has a sesamoid bone or fabella that articulates with a femoral condyle	Dorsal surface of the calcaneal tuberosity by the common calcaneal tendon to provide the largest portion of the tendon		Tibial nerve
Superficial digital flexor	Flexor digitorum brevis	MTP and PIP joints of II-V; stifle flexion, talocrural extension	Lateral supracondylar tuberosity of the femur	Calcaneal tuberosity and bases of middle phalanges digits II-V		Tibial nerve
Deep digital flexor	Flexor digitorum longus	Digit flexion; talocrural extension by both parts	See each part separately below	Two heads join at the distal tarsals and then attach to the plantar surface of phalanges I-V	Distomedial the base of distal	Tibial nerve
Lateral digital flexor (flexor hallucis longus)	Flexor hallucis longus		Caudolateral border of the proximal two thirds of tibia, proximal half of fibula, and adjacent interosseous membrane	Before joining the medial digital flexor muscle, the tendon is bound in a groove at the sustentaculum tali		
Medial digital flexor (flexor digitorum longus)	Flexor digitorum longus		Caudomedial border of the tibia, popliteal line head of the fibula	Tendon lies on the caudomedial side of the tibia before joining the lateral digital flexor muscle		

Caudal tibial	Posterior tibialis/ tibialis posterior	Rotation of the hindpaw so the plantar paw faces medially; talocrural extension	Medial proximal fibula	Medial ligamentous masses of the tarsus after coursing as a very thin tendon, lying cranial to medial digital flexor at the talocrural joint	Distomedial	Tibial nerve
DEEP DORSAL MUSCLES OF THE NECK						
Splenius	Splenius	Extension, ipsilateral side bend, and ipsilateral rotation of the neck, raising the head and neck, stabilization of T1	Spinous processes of T1-T2, ligamentum nuchae cranial to T1, median dorsal raphe of cervical spine, by aponeurosis of cranial part of thoracolumbar fascia, extending to T5-T6	Mastoid part of the temporal bone with the longissimus capitis, dorsal nuchal line of occiput, at times to the wing of the atlas	Cranioventral	Dorsal rami of cervical nerves
DEEP VENTRAL MUSCLES OF THE NECK						
Scalenus: two parts	Scalenus	Ipsilateral side bend of the neck, pulls the neck down, inspiration				
Supracostal scalenus— superficial part (two or three parts)			Lateral surfaces of ribs 1-9	Transverse processes of C4 and C5 and sometimes C3	Cranio-medio- dorsal	Ventral rami of cervical and thoracic spinal nerves
Præcostae scalenus—deep part (three parts)			As three portions from the cranial border of rib 1	Superficial part to transverse process of C3 through C5; two deep parts to transverse processes of C6 and C7	Cranio-medio- dorsal	
Longus capitis	Longus capitis	AO flexion, pulls the neck down	Caudal surfaces of the transverse processes of cervical vertebrae	Ventral surface of the basioccipital bone of the skull	Cranial	Ventral rami of cervical spinal nerves

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
Longus colli	Longus colli	Neck flexion, pulls the neck down	Ventral convex surfaces of T1-T6 vertebrae Ventral borders of transverse processes of C6-C3 Ventral surface of the angle of cranial ribs	Ventral border of the wing of C6 and the transverse process of C7 Ventral aspect of the next cranial vertebra Ventral aspect of the atlas	Cranial Craniomedial Craniomedial	Ventral rami of cervical spinal nerves
LATERAL AND VENTRAL THORACIC WALL MUSCLES						
External intercostals	External intercostals	Draw ribs together; breathing	Caudal border of a rib	Cranial border of the next caudal rib	Caudovertral	Intercostal nerves
Internal intercostals	Internal intercostals	Draw ribs together; breathing	Extending from the vertebral column to the distal ends of the rib—cranial border of a rib	Caudal border of the next cranial rib	Craniovertral	Intercostal nerves
MUSCLES OF THE ABDOMINAL WALL AND DIAPHRAGM						
Rectus abdominis	Rectus abdominis	Flexion of the spine, bring pelvis forward (i.e., posterior pelvic tilt); abdominal press functions—expiration, urination, defecation, parturition; support of abdominal viscera	Sternum, rib 1, and 1st costal cartilage, and a fleshy attachment on the sternal portion of the 9th costal cartilage	Pecten ossis pubis, between the two iliopectineal eminences	Caudal	Intercostal, costoabdominal, ilioinguinal nerves, medial branches
External abdominal oblique	External oblique	Flexion of the spine, ipsilateral side bend, functions of abdominal press, contralateral rotation of the spine, support of abdominal viscera	Last ribs, thorocolumbar fascia	Linea alba, cranial pubic ligament	Caudovertral	Intercostal nerves 3 or 4-12, costoabdominal, iliohypogastric and ilioinguinal nerves, lateral branches

Internal abdominal oblique: abdominis three parts	Internal oblique abdominis	Flexion of the spine, ipsilateral side bend, functions of abdominal press, support of abdominal viscera	Thoracolumbar fascia caudal to rib 12 with the pars lumbalis	Rectus abdominis and linea alba	Cranioventral	Caudal intercostal nerve and medial branches of the costoabdominal, iliohypogastric, and ilioinguinal nerves
Transversus abdominis	Transversus abdominis	Abdominal press, support of abdominal viscera	Costal cartilage of rib 8, transverse process of L7, and tuber coxae	Linea alba	Ventral	Caudal intercostal nerve and medial branches of the costoabdominal nerve
Diaphragm	Diaphragm	Inspiration, support of abdominal viscera	Ventral surfaces of lumbar vertebrae, ribs, and sternum	Central tendon	Craniomedial	Phrenic nerve, ventral rami of C5-C7
DEEP DORSAL MUSCLES OF THE NECK AND TRUNK						
Iliocostalis system: two parts	Iliocostalis	Ipsilateral side bend; stabilizes the spine, expiration by pulling the ribs caudally				Dorsal rami of thoracic and lumbar nerves
Iliocostalis thoracis	Iliocostalis thoracis		Cranial borders of vertebral ends of ribs 2-12	Costal angles of the ribs, transverse process of C7	Cranial	
Iliocostalis lumborum	Iliocostalis lumborum		Pelvic surface of the wing of the ilium, the iliac crest, and the intermuscular septum between the iliocostalis and longissimus muscles	Lumbar vertebrae and last 5 ribs	Cranioventral	
Longissimus thoracis et lumborum: two parts—strongest muscle of the trunk in the thoracic and lumbar region	Longissimus thoracis et lumborum	Extension of the spine, raising the cranial portion of the body from the pelvis; stabilizes the spine, raises the caudal body with the hindlimbs			Craniolateral	Dorsal rami of thoracic and lumbar nerves

LONGISSIMUS SYSTEM: THREE PARTS—EXTENT IS FROM THE ILIAC CREST TO C7

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
Longissimus thoracis	Longissimus thoracis		Spinous processes and supraspinous ligament of thoracic vertebrae	Medial tendons to the accessory processes of T6-T13 and caudal ends of transverse processes of T1-T5 Lateral tendons to the groove adjacent to the costal tubercle of ribs 6-13 Cranially, both attach to the costal tubercles of ribs 1-5		
Longissimus lumborum	Longissimus lumborum		Iliac crest and ventral surface of the ilium, supraspinous ligament of lumbar vertebrae	Dorsal aspects and cranial articular processes of lumbar vertebrae, disk of lumbosacral joint		
Longissimus cervicis	Longissimus cervicis	Extension of the neck, ipsilateral side bend and rotation	Spinous processes and supraspinous ligament of cranial thoracic and caudal cervical vertebrae	Transverse processes of cranial cervical vertebrae		Dorsal rami of cervical and thoracic nerves
Longissimus capitis	Longissimus capitis	AO extension, ipsilateral side bend and rotation	Transverse processes of T1-T3, caudal articular processes of C3-C7	Mastoid process of the temporal bone, united with splenius at the level of the atlas		Dorsal rami of cervical nerves
TRANSVERSOSPINALIS SYSTEM—JOINS ONE OR MORE VERTEBRAE						
	Transversus spinalis					Dorsal rami

Spinalis et semispinalis thoracis et cervicis: two parts	Spinalis and semispinalis thoracis	Neck extension, raises the neck, stabilizes the spine	Dorsal aspect of the longissimus thoracis muscle, spinous processes and accessory processes of cranial lumbar vertebrae and C7-T6 and mammillary processes of T13-L2, iliac crest, spinous processes of T13-S2	Thoracis portion—spinous processes of thoracic vertebrae Cervicis portion—spinous processes of C2-C5 vertebrae	Craniomedial	Dorsal rami of thoracic nerves Dorsal rami of cervical nerves
Spinalis et semispinalis thoracis—lateral						
Spinalis cervicis—medial	Spinalis cervicis					
Semispinalis capitis: two parts	Semispinalis capitis	Neck extension, ipsilateral side bend, stabilize the neck				Dorsal rami of cervical nerves
Multifidus: three parts	Multifidus	Stabilize the vertebral column			Craniomedial	Dorsal rami of cervical, thoracic, and lumbar nerves Dorsal rami of lumbar and thoracic nerves
Multifidus lumborum	Multifidus lumborum		Dorsal sacrum, mammillary processes of L7-T12	Spinous processes of T9-L6		
Multifidus thoracis	Multifidus thoracis		Mammillary and transverse processes of T3-T11	Spinous processes of T7-T8		Dorsal rami of thoracic nerves
Multifidus cervicis	Multifidus cervicis		Articular process of T2	Spinous process of axis		Dorsal rami of cervical nerves

Continued

Muscle	Correlated Human Muscle	Joint Action in the Forelimb	Main Cranial, Dorsal, or Proximal Attachment	Main Caudal, Ventral, or Distal Attachment	Fiber Direction from Proximal to Distal Attachment (listed in order of importance)	Innervation
MUSCLES OF THE TAIL						
Sacrocaudalis dorsalis lateralis	None	Tail extension or lifting, ipsilateral side bend	Aponeurosis of the longissimus, mammillary processes L1-L6 and Cd1-Cd8 articular processes of the sacrum	Mammillary processes of Cd vertebrae Continuation of the longissimus muscle	Caudomedial	Dorsal rami of caudal nerves
Sacrocaudalis dorsalis medialis	None	Tail extension or lifting, ipsilateral side bend	Spine and dorsolateral aspects of cranial vertebrae	Dorsolateral aspects of caudal vertebrae	Caudolateral	Dorsal rami of caudal nerves
Sacrocaudalis ventralis lateralis	None	Tail flexion, ipsilateral side bend	Ventral surface of L7 and the sacrum, ventral surfaces of transverse processes of Cd vertebrae	Ventrolateral tubercle of more caudal vertebrae	Caudomedial	Ventral rami of caudal nerves
Sacrocaudalis ventralis medialis	None	Tail flexion, ipsilateral side bend	Ventral surface of S3 and Cd vertebrae	Hemal process of the next caudal vertebrae	Caudomedial	Ventral rami of caudal nerves

AA, Atlantoaxial; AO, atlantooccipital; caudal *DIS*, caudal dorsal iliac spine; *C*, cervical spinal nerve; *caudal V/S*, caudal ventral iliac spine; *Cd*, caudal; *CN*, cranial nerve; *cranial DIS*, cranial dorsal iliac spine; *cranial V/S*, cranial ventral iliac spine; *Cx*, coccygeal; *L*, lumbar spinal nerve; *MC*, metacarpal bone; *MCP*, metacarpophalangeal joint; *MT*, metatarsal bone; *MTP*, metatarsophalangeal joint; *PIP/DIP*, proximal/distal interphalangeal joint; *S*, sacral spinal joint.



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WEB APPENDIX

Main Ligament Attachments and Function

Name of Ligament	Correlated Human Ligament	Proximal Attachment	Distal Attachment	Fiber Direction	Motion Restricted by Ligament/Function
FORELIMB SHOULDER JOINT					
Transverse humeral retinaculum	Transverse humeral	Medial side of greater tubercle	Lateral side of lesser tubercle	Transverse/horizontal	Stabilizes the biceps brachii, proximal tendon
Medial glenohumeral	Glenohumeral	Medial lip of glenoid fossa	Greater tubercle	Distal	Stabilizes humeral head, preventing medial dislocation
Lateral glenohumeral	Glenohumeral	Lateral lip of glenoid fossa	Lesser tubercle	Distolateral	Stabilizes humeral head, preventing lateral dislocation
ELBOW JOINT					
Medial collateral	Lateral or radial collateral	Humeral medial epicondyle	Radial tuberosity and radius and ulna in the interosseous space by crura	Distal	Limits elbow valgus
Lateral collateral	Medial or ulnar collateral	Humeral lateral epicondyle	Ulna with the annular ligament and area distal to the radial neck	Distal	Limits elbow varus
RADIOULNAR JOINTS					
Annular	Annular	Lateral extremity of the radial notch of the ulna	Medial extremity of the radial notch of the ulna	Circular	Stabilizes radial head position
Interosseous	Interosseous	Radius	Ulna	Between ulna and radius	Stabilizes radius and ulna in radioulnar pronation
Radioulnar	Distal radioulnar	Distal, lateral radius	Distal, medial ulna	Transverse	Stabilize the distal radioulnar joint
CARPAL JOINTS					
Flexor retinaculum/transverse palmar carpal	Transverse carpal	Accessory carpal and styloid process of radius	Palmar projections of radius and proximal carpals	Transverse	Two layers stabilize tendons of the superficial and deep digital flexors
Short radial collateral	Radial collateral of the wrist	Tubercle proximal to radial styloid process	Medial radial carpal	Distal	Valgus deviation
Straight part		Radial styloid process	Palmomedial surface of radial carpal	Distal	Valgus deviation
Oblique part					
Short ulnar collateral	Ulnar or medial collateral	Ulnar styloid process	Ulnar carpal	Distal	Varus deviation

Dorsal radiocarpal	Radial or lateral collateral	Dorsolateral radius	Dorsoproximal ulnar carpal	Distolateral	Carpal flexion
Palmar carpal	Palmar carpal ligaments	Palmar aspect of radius, ulna, and carpals	Distally on the carpals and metacarpals	Distal, distolateral, distomedial	Carpal extension
DIGITAL JOINTS					
Collaterals of the MCP joints	Collaterals of the MCP joints	Distal ends of metacarpals	Proximal ends of proximal phalanges	Vertical in weight-bearing	Medial and lateral MCP joint motion
Collaterals of the PIP joints	Collaterals of the PIP joints	Distal ends of proximal phalanges	Collateral tubercles on the proximal ends of the middle phalanges and distal phalanges in digit I	Vertical in weight-bearing	Medial and lateral PIP motion
Collaterals of the DIP joints	Collaterals of the DIP joints	Distal ends of middle phalanges	Sides of the ungual crests of the distal phalanx	Caudodistal	Medial and lateral PIP motion
HINDLIMB HIP JOINT					
Transverse acetabular	Transverse acetabular	One side of acetabular notch	Other side of acetabular notch	Caudal	Completes the acetabular lip to secure femoral head in the acetabulum
Ligament of the femoral head	Ligamentum teres or ligamentum capitis femoris	Fovea in the femoral head	Acetabular fossa	Mediocranial	Maintains hip reduction
STIFLE JOINT					
Cranial cruciate	Anterior cruciate	Intercondylar fossa, caudomedial part of the femoral lateral condyle	Cranial intercondylar area of the tibia	Craniodistal	Stifle cranial glide during extension, also internal rotation
Caudal cruciate	Posterior cruciate	Intercondylar fossa, femoral medial condyle	Medial edge, popliteal notch of tibia, caudal to the caudal attachment of the medial meniscus	Caudodistal	Stifle caudal glide during flexion
Medial collateral	Medial or tibial collateral	Medial femoral epicondyle	Medial tibial border, distal to the condyle	Distocranial	Stifle valgus and extension
Lateral collateral	Lateral or fibular collateral	Lateral femoral epicondyle	Attaches to the medial meniscus		
Patellar ligament	Patellar ligament (patellar tendon)	Quadriceps tendon and patella	Fibular head and adjacent tibial condyle	Distal (vertical instance)	Stifle varus and extension
			Tibial tuberosity	Distal	Transfer force of quadriceps muscle to the tibia

Continued

Name of Ligament	Correlated Human Ligament	Proximal Attachment	Distal Attachment	Fiber Direction	Motion Restricted by Ligament/Function
TALOCURRAL JOINT					
Medial collateral	Medial or tibial collateral				
Long part		Medial aspect of the medial malleolus	Firmly to tarsal I and the central tarsal and loosely to MTs I-II and the talus	Distocranial	Stabilization of lateral motion of the tarsus
Short part: two parts		Medial aspect of the medial malleolus	1. Tarsal I and MTs, sustentaculum tali (deep to long part) 2. Talus	Distocranial Caudal	
Lateral collateral	Lateral or fibular collateral				Stabilization of internal motion of the tarsus
Long part		Lateral malleolus	Calcaneus, tarsal IV, and base of MT V	Distocranial	
Short part: two parts		Lateral malleolus	1. Tuber calcanei 2. Talus	Distocaudal Caudodistal	
TARSUS					
Distal extensor—retinaculum—distal transverse ligament of tarsus	None	Calcaneus	Surrounds the long digital extensor to attach to the calcaneus	Circular	Secure the long digital extensor muscle
Dorsal ligaments—many, named by connecting bones	Many dorsal ligaments	From one tarsal bone	Another tarsal bone or MT bone	Different for each ligament	Stabilize the dorsal tarsus Limit tarsal extension
Talocalcaneal	Subtalar	Talus—distal surface	Calcaneus—proximal surface	Near vertical	Stabilize the talocalcaneal articulation
Plantar ligaments—thicker and stronger than dorsal talocrural ligaments	Long and short plantar ligaments				Support the joints of the plantar hindpaw Increase stability Limit dorsiflexion
Medial plantar		Sustentaculum tali, plantar surface	Central tarsal and tarsometatarsal joint capsule	Distolateral	

Middle plantar—most commonly injured plantar ligament	Body of the calcaneus, plantar aspect	Tarsal bone IV and bases of MTs IV and V	Distal	
Lateral plantar	Caudolateral calcaneus	Long part of lateral collateral ligament of the talocrural joint and base of MT V	Distal	
DIGITAL JOINTS				
Collaterals of the MTP joints	Distal ends of MTs	Proximal ends of proximal phalanges	Vertical in weight-bearing	Abduction and adduction MTP motion
Collaterals of the PIP joints	Distal ends of proximal phalanges	Collateral tubercles on the proximal ends of the middle phalanges and distal phalanx in digit I	Vertical in weight-bearing	Abduction and adduction PIP motion
Collaterals of the DIP joints	Distal ends of middle phalanges	Sides of the ungual crests of the distal phalanx	Caudodistal	Abduction and adduction PIP motion
Proximal and distal digital annular	One side of phalanx	Other side of phalanx crossing the plantar surface	Mediolateral	Secure the flexor tendons close to the phalanges
SPINE				
AO Joint				
Dorsal AO ligament or membrane	Dorsal edge of foramen magnum	Cranial border of the arch of the atlas	Craniocaudal	AO flexion Punctures are made through this membrane to drain cerebrospinal fluid AO extension
Ventral AO ligament or membrane	Ventral edge of foramen magnum	Body of the atlas	Craniocaudal	
Lateral AO ligament	Lateral part of the dorsal arch of the atlas	Jugular process of the occipital bone	Craniioventro-lateral	AO contralateral side bend
AA Joint				
Dorsal AA membrane	Caudal margins of the sides and body of the atlas	Cranial margin of the same areas of the axis	Craniocaudal	AA contralateral side bend
Apical ligament of the dens	Dens of the axis	Foramen magnum, occipital bone medial to the caudal parts of the occipital condyles	Cranial and craniolateral	Stabilizes the dens
Transverse ligament of the atlas	Internal surface of the ventral arch of the atlas	Internal surface of the other side of the arch of the atlas	Mediolateral	Stabilizes the dens

Correlated Human		Proximal Attachment	Distal Attachment	Fiber Direction	Motion Restricted by Ligament/Function
Name of Ligament	Ligament				
INTERVERTEBRAL JOINTS					
Ligamentum nuchae/nuchal ligament	Ligamentum nuchae	Caudal part of the spinous process of C2	Tip of spinous process of T1	Craniocaudal	Stabilizes the cervical spine
Supraspinous	Supraspinous	Spinous process, T1 being the most cranial	Each sequential spinous process to Cd3	Craniocaudal	Limits separation of the spinous processes during spine flexion motion
Ventral longitudinal	Anterior longitudinal	Ventral surfaces of bodies of vertebrae, C2 being the most cranial Best developed caudal to the middle thorax	Ventral surfaces of bodies of vertebrae, the sacrum being the most caudal	Craniocaudal	Limits spine extension
Dorsal longitudinal	Posterior longitudinal	Dorsal surfaces of bodies of vertebrae, C2 being the most cranial Heavier than the ventral longitudinal ligament	Dorsal surfaces of bodies of vertebrae, the end of the vertebral canal in the Cd region being the most caudal	Craniocaudal	Limits spine flexion
Interspinous	Interspinous	Cranial border of a spinous process	Caudal border of a spinous process	Craniocaudal	Limits spine flexion
Intertransverse	Intertransverse	Transverse process of a caudal vertebra	Transverse process of an adjacent cranial vertebra	Craniolateral	Limits side bend
Yellow/interarcuate/ligamentum flavum	Ligamentum flavum	Loose thin elastic ligament attached to the arch of a vertebra	Arch of an adjacent vertebra	Craniocaudal	Limits spine flexion

LIGAMENTS OF THE RIBS

Ligament of the head	Head of the rib	Lateral part of the intervertebral disk and body adjacent to the disk	Mediolateral	Stabilizes the rib
Ligament of the tubercle/dorsal costotransverse ligament	Distal to the articular capsule of the tubercle	Transverse process of the vertebra corresponding to the rib	Mediolateral	Stabilizes the rib
Ligament of the neck	Neck of the rib	Ventral surface of transverse process and adjacent lateral surface of the body of the vertebra	Medioventral	Stabilizes the rib

SACROILIAC JOINT

Ventral sacroiliac	Ventral sacroiliac	Ilium, ventral surface	From the sacrum, dorsolateral and medial	Stabilizes the sacroiliac joint, prevents ventral displacement of the sacrum
Dorsal sacroiliac	Caudal dorsal iliac spine	Caudal two thirds of the lateral border of the sacrum or even Cd1	Caudomedial portion and dorsocaudal portion	Stabilizes the sacroiliac joint
Sacrospinous	Caudolateral apex of the sacrum and Cd1	Lateral angle of the ischiatic tuberosity	Caudolatero-ventral	Stabilizes the sacrum and pelvis Muscle attachment site

AA, Atlantoaxial; AQ, atlantooccipital; Cd, caudal; DIP, distal interphalangeal; L, lumbar; MCP, metacarpophalangeal; MTP, metatarsophalangeal; PIP, proximal interphalangeal; S, sacral.



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WEB APPENDIX

Manufacturers of Supplies and Equipment

Aquatic Therapy

Aqua Paws Underwater Treadmill System Platinum
http://www.fernovetsystems.com/smallanimal/ap_2chamber.htm

1 (888) 206-7802

Canine Aquatherapy (Underwater Treadmills, Pools)
<http://www.canineaquatherapy.com/main/Main/main.html>
1 (604) 888-2525

Hydro Physio
<http://www.hydrophysio.com>
1 (866) 385-7676

Water Walker Treadmill (underwater treadmill)
http://www.animalrehab.co.uk/animalrehab/Water_Walker/
1 (877) 426-9255

Hydro Hound
<http://www.hydrohound.com/>
1 (541) 798-0110

Paws Aboard Life Jackets
<http://www.pawsaboard.com/dog-lifejacket-pet-life-jacket.asp>
1 (317) 222-1112

Hydro Horse
<http://horsetreadmills.com/>
1 (800) 405-1099

Physio-Tech Aqua Treadmill
<http://www.physio-treadmill.egah.de/>
+49 40232384

Aquatic Access Inc. (dog lift for pools)
www.aquaticaccess.com
1 (502) 425-5817

VetSystems Whirlpool
<http://www.vetsystems.net>
1 (843) 572-0708

Cryotherapy

Canine Icers
<http://www.canineicer.com>
1 (434) 975-5939

Grampa's Garden
<http://www.grampasgarden.com/>
1 (207) 373-0770

Canine Chillows
<http://www.eoco.org.uk/chillowspets.shtml>
+44 1256 332737

Pro-Kold Ice Wraps
<http://www.equinetherapyproducts.com/store/product/4>
1 (800) 425-9506

Ice Horse
<http://www.mackinnonicehorse.com/>
1 (800) 786-6633

Game Ready Vasopneumatic Compressive Devices
<http://www.gamereadyvet.com/>
1 (888) 426-3732

DVDs/Books

Building the Canine Athlete
<http://www.caninesports.com>
1 (443) 535-9144

Wizards of Paws
<http://www.wizardofpaws.net>
1 (860) 537-1081

Long Term Care Pets
<http://www.activitytherapy.com/pet.htm>
1 (973) 729-6601

Sound Therapy
<http://www.soundtherapy4pets.com/pet-therapy.php>
+44 1244 371 473

Pet Therapy Services
<http://www.pettherapyservices.org/products.html>
chris@pettherapyservices.org

Electrical Stimulation

Intellect Vet Electrotherapy System
http://www.fernovetsystems.com/equine/es_intelect.htm
1 (888) 206-7802

Matrix Therapy Products
<http://therapyproducts.net/>
1 (503) 632-7187

Alpha Stim SCS
www.midwestmicrocurrent.com/vetindex.html
1 (888) 583-0944

Kits/Equipment

Canine Rehab Products
<http://www.caninerehabproducts.com>
1 (413) 243-0253

Balanced Canine Products
<http://www.balancedcanineproducts.com>
1 (954) 655-2584

Animal Therapy Systems
<http://www.animaltherapysystems.com/products/>
1 (800) 544-8957

Jog A Dog Treadmill
<http://www.jogadog.com/>
1 (800) 756-4364

FitPAWS Canine Conditioning Equipment
<http://www.ballodynamics.com/fitpaws.php>
1 (800) 752-2255

DogTread Treadmill
http://www.petzenproducts.com/health_d353.htm
1 (801) 866-3955

Lasers

K Laser
<http://www.k-laserusa.com/your-k-laser/animal-veterinary/>
1 (877) 776-8246

Low Level Laser Therapy
<http://www.respondsystems.com>
1 (800) 722-1228

Low Level Laser Therapy System
<http://www.danscottandassociates.com>
1 (888) 866-6736

Pulsing Magnetic Therapy
<http://www.respondsystems.com>
1 (800) 722-1228

Emerson Worldwide (lasers, infrared, magnetics, etc.)
<http://www.emersonww.com/>
1 (888) 974-5687

Companion Therapy Laser
www.companiontherapy.com
1 (302) 709-0408

SpectraVET Therapeutic Lasers
www.spectravet.com
1 (704) 263-8101

Irradia Education & Resources (laser therapy education)
www.irradia.us
1 (704) 263-5525

Splints, Braces, Orthotics, Prosthetics, and Supports

Animal Ortho Care, LLC
<http://www.animalorthocare.com>
1 (703) 474-6204

A-Trac Dynamic Brace
<http://www.woundwear.com>
1 (800) 443-4055

Dynasplint
<http://www.dynasplint.com>
1 (800) 638-6771

Jorvet Pet Brace
<http://www.jorvet.com>
1 (800) 525-5614

Orthovet Splints
<http://www.orthovet.com>
1 (541) 544-2435

Tammy and Teddy's Booties and More
<http://tammyandteddys.com>
1 (860) 749-6552

Thera-Paw Pet Products
<http://www.therapaw.com>
1 (800) 525-5614

Canine Orthopedic Leg Braces
<http://www.dogbrace.com/>
1 (209) 688-4047

Bottoms Up Leash
<http://www.bottomsupleash.com/>
1 (800) 805-2001

Walkabout Harnesses
<http://www.walkaboutharnesses.com/>
1 (800) 779-0439

Ace Ortho Solutions
<http://www.aceorthosolutions.com>
1 (308) 423-2690

Help 'Em Up Harness
<http://www.helpemup.com/>
1 (303) 447-2201

K-9 Orthotics & Prosthetics, Inc.

www.k-9orthotics.com

1 (902) 865-5596

Ortho Pet Custom Prosthetics and Orthotics

<http://www.orthopets.com/>

1 (303) 953-2545

Sky Prosthetics, Inc.

<http://www.skyprosthetics.com>

1 (866) 832-1269

Ortho Dog Mobility Braces

<http://www.orthodog.com/>

1 (828) 298-0885

Biko Canine Physio Brace

<http://www.biko.co.at/>

<http://www.biko.co.at/english/default.html>

+43 699 15200080

Thermotherapy

Infrared Therapeutic Pet Pads and Beds

<http://www.mtavet.com>

1 (866) 311-1303

Therapeutic Pet Beds

<http://www.thermotexusa.com/products-pets.htm>

1 (513) 942-2770

Wheelchairs

Doggon' Wheels

<http://www.doggon.com>

1 (800) 736-4466

Eddie's Wheels for Pets

<http://www.eddieswheels.com>

1 (888) 211-2700

Tammy and Teddy's Booties and more

<http://tammyandteddys.com>

1 (860) 749-6552

Treadmills, wheelchairs, etc.

<http://ablepet.com/Therapeutic-Exercise/CanineRehabProducts.html>

1 (413) 243-0253

K9 carts

<http://www.k9-carts.com/>

1 (800) 578-6960

Animal Therapy Products

<http://www.animaltherapyproducts.com/>

1 (904) 779-2590

Dewey's Wheelchairs for Dogs

<http://www.wheelchairsfordogs.com/>

1 (877) 312-2122

Walkin' Wheels Wheelchairs for Dogs

<http://www.walkinwheels.com/>

1 (603) 577-8854

Canine Wheelchair Project

web.mac.com/gbertocci

canine.biomechanics@gmail.com

Wraps (Incontinence) & Modified Items

Sam's Dog Hut

<http://www.samsdoghut.com>

1 (888) 908-4888

DogLeggs

<http://www.dogleggs.com>

1 (800) 313-1218

Advanced Canine

<http://www.dogpt.com/catalog.htm>

1 (908) 447-3876

Hot or Cold Leg Wraps

<http://www.animaltherapysystems.com/rejuvangelz/>

1 (800) 544-8957

The Femwrap (for urinary incontinence)

<http://www.arwi.com/products.html>

1 (919) 861-5868

Rubber Dog Boots

<http://www.pawzdogboots.com>

1 (646) 301-8959

Pet Therapy Pads & Pillows

<http://www.pettherapypadsandpillows.com/productpage.htm>

1-866-595-7749

gonerennie@aol.com

Puppy Stairs

<http://www.puppystairs.com>

1 (866) 767-4551

Medipaw Protective Boot

<http://www.medivetproducts.com/>

1 (800) 245-3413

The Anxiety Wrap

<http://www.anxietywrap.com/>

1 (260) 344-1217

Miscellaneous

Magnetic Therapy

<http://www.therionmagnetics.com/magnetic-pet-products.html>

1 (888) 636-9700

Horse Magnetic Equipment

<http://www.equinemagnetic.com/Pet.html>

1 (800) 731-8463

Magnetic Pet Bed

<http://magnapak-magnetic-mattress-health-therapy.com/catalog/index.php>

1 (519) 471-8611

Ramps, Beds, Vests

<http://ablepet.com/Therapeutic-Exercise/CanineRehabProducts.html>

+61 (3) 9793 6188

Shampoo for Dog Skin Conditions

<http://www.ravenpetproducts.com/>

1 (800) 888-3647

Equineox Technologies (hyperbaric chamber)

<http://www.equineox.com/index2.html>

1 (604) 465-3134

Equine Therapy Products

<http://www.equinetherapyproducts.com/>

1 (800) 425-9506

Equine Health Care

<http://www.equinehealthcare.com/>

1 (859) 223-1326

Handicapped Pets

<http://handicappedpets.com/www/index.php/pet-products-summary.html>

1 (888) 811-7387

Senior Pet Products

<http://www.seniorpetproducts.com/>

1 (800) 523-7979

Scout's Store ("special needs" pet products)

<http://www.scoutshouse.com/store/>

1 (800) 558-8951

FitPAWS Canine Conditioning Equipment

<http://www.ballodynamics.com/fitpaws.php>

1 (800) 752-2255

Lite Gait Mobility Training

<http://litegait.com/litegaitv.html>

1 (800) 332-9255

GAITRite Gait Analysis Software

<http://gaitfour.com/site/>

1 (610) 449-4879

Motion Imaging and Analysis

<http://www.mi-as.com/>

1 (805) 577-0463

Ruff Wear Performance Dog Gear

www.ruffwear.com

1 (888) 783-3932